NATIONAL CHILDREN'S SCIENCE CONGRESS (NCSC) 2026–27

KRISHI-DRISHTI: TECHNICAL EVIDENCE & SUPPORTING ANNEXURE

Document 03: Technical Blueprints, Schematics, Verification Tables & Scientific Data

Document Purpose: *This technical annexure serves as the primary supporting reference for the Main Project Report. It provides verified electrical pinout tables, communications state-machine contracts, mathematical gating formulations, cryptographic dataset audits, complete empirical benchmark matrices, and a plain-language technical glossary. All data and pinouts are directly extracted from the compiled firmware and tested software in the project repository.*

SECTION 1: HARDWARE WIRING, INTERFACES & PINOUT TABLE

Verified directly against KRISHI_DRISHTI_Controller.ino and KRISHI_DRISHTI_Camera.ino:

Subsystem	Component Pin	ESP32 Microcontroller GPIO	Signal Type / Electrical Specification	Firmware Configuration / Pull-up
Controller Wand	SSD1306 SDA	GPIO 21	I2C Serial Data (3.3V)	Internal Pullup / 400kHz Fast Mode
Controller Wand	SSD1306 SCL	GPIO 22	I2C Serial Clock (3.3V)	Internal Pullup / 400kHz Fast Mode
Controller Wand	SCAN Button	GPIO 27	Active LOW momentary switch	INPUT_PULLUP, software debounced 50ms
Controller Wand	NEXT Button	GPIO 26	Active LOW momentary switch	INPUT_PULLUP, software debounced 50ms
Controller Wand	DHT11 DATA	GPIO 18	Single-wire bi-directional digital	Bit-banged timing, 10k external pullup
Controller Wand	Power In (VCC)	5V / VIN pin	Regulated 5.0V DC from power bank	Filtered via 100uF capacitor
Camera Unit	Flash Torch LED	GPIO 4	High-power onboard white LED	PWM controlled / active HIGH trigger
Camera Unit	OV5640 SIOD (SDA)	GPIO 26	SCCB Serial Data interface	Hardware I2C bridge to camera sensor
Camera Unit	OV5640 SIOC (SCL)	GPIO 27	SCCB Serial Clock interface	Hardware I2C bridge to camera sensor
Camera Unit	OV5640 XCLK	GPIO 0	Master Clock output (20 MHz)	LEDC hardware timer square wave
Camera Unit	MicroSD D0 / CLK	GPIO 2 / GPIO 14	SD_MMC 1-Bit high speed bus	DMA accelerated block transfer (512B)

SECTION 2: HAND-DRAWABLE CIRCUIT SCHEMATIC (ASCII LAYOUT)

| VCC -> 3.3V VCC -> 3.3V PIN -> 27 PIN -> 26 |
| GND -> GND GND -> GND GND -> GND GND -> GND |
| SDA -> GPIO 21 DATA -> GPIO 18 |
| SCL -> GPIO 22 (10k pullup to 3.3V) |
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|| Wi-Fi 802.11 b/g/n (SoftAP @ 192.168.4.1)
+----------+
| AI-THINKER ESP32-CAM UNIT |
| 5V In <■■ Power Bank | GND <■■ Common GND |
| OV5640 5MP Sensor <■■ DVP 8-bit bus (GPIOs 35,34,39,36,21,19,18,5) |
| Flash LED <■■ GPIO 4 (Diffuse acrylic disc) |
| MicroSD Card <■■ SD_MMC 1-Bit mode (GPIOs 2, 14, 15) |
| Embedded PROGMEM HTTP Server: Port 80 (Endpoints: /api/capture, /api/diagnosis)|
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SECTION 3: NETWORK REST API CONTRACTS & COMMUNICATIONS FLOW

KRISHI-DRISHTI uses a decentralized, lightweight HTTP REST architecture over local Wi-Fi SoftAP (SSID: KRISHI-DRISHTI, Password: 12345678, IP: 192.168.4.1). No router, cellular tower, or external internet gateway is ever required.

API Endpoint	Method	Initiator & Receiver	Payload / Query Parameters	System Function & Response
/api/status	GET	Wand ■■> Camera	None	Returns JSON with cameraReady, sdReady, scanCounter.
/api/capture	GET	Wand ■■> Camera	None	Triggers OV5640 capture, writes JPEG to SD card, increments scan_id.
/api/telemetry/set	GET	Wand ■■> Camera	?temp=22.4&hum;=88.5	Buffers instantaneous canopy temperature and humidity in camera memory.
/api/diagnosis/set	GET	Phone ■■> Camera	?crop=Tomato&disease;=LB&conf;=99.5&status;=SEVERE	Smartphone AI daemon uploads neural prediction and agronomic advice.
/api/diagnosis/latest	GET	Wand ■■> Camera	None	Wand polls every 250ms until diagnosisReady == true, then updates OLED.

SECTION 4: OFFLINE AI PIPELINE & DUAL-STAGE GATING FORMULATIONS

The AI inference daemon (krishi_ai_daemon.py) executes entirely on the farmer's smartphone. It employs mathematical checks to prevent misleading diagnoses on corrupted, blurry, or non-foliar frames:

1. Foliar Plausibility Ratio (R_{foliar}):
Each pixel (r, g, b) is evaluated for healthy chlorophyll ($g > r$ and $g > b$) or necrotic lesion pigmentation ($r > 60, g > 40, b < 60$).
$$R_{\text{foliar}} = \frac{N_{\text{green}} + N_{\text{necrotic}}}{N_{\text{total}}} \times 100\%$$
If $R_{\text{foliar}} < 12.0\%$, the frame is immediately rejected as NON_LEAF (bare soil, mulch, clothing, or hands).

2. Peripheral Dark Void Gate (V_{dark}):
Measures the percentage of perimeter border pixels with brightness < 35 (0–255 scale).
$$V_{\text{dark}} = \frac{N_{\text{border}}(\text{luminance} < 35)}{N_{\text{border total}}} \times 100\%$$
If $V_{\text{dark}} > 45.0\%$, the image is flagged for dark void contamination (intercepting black background cloths and flashlight fall-offs).

3. Shannon Information Entropy (H):
Measures prediction ambiguity across the 6 discrete output softmax probabilities $p_1, p_2, \dots, p_6$:
$$H = -\sum_{i=1}^{6} p_i \ln(p_i) \quad [\text{in nats}]$$
• A confident prediction (e.g., $p = [0.99, 0.002, 0.002, 0.002, 0.002, 0.002]$) yields $H \approx 0.06$ nats.
• A confused, uniform distribution ($p_i \approx 0.167$) yields $H = \ln(6) \approx 1.79$ nats.
• **Gating Criterion:** If $H > 1.45$ nats OR $p_{\max} < 0.50$, the prediction is triaged as UNCERTAIN, preventing false confidence on ambiguous frames.

SECTION 5: DATASET PROVENANCE & CRYPTOGRAPHIC DEDUPLICATION AUDIT

To establish reproducible zero-leakage benchmarks, I conducted an independent SHA-256 hash audit documented in ai_engine/TRAINING_RUN_MANIFEST.json.

• **Source Sampling:** 6,600 PlantVillage image files (1,100 per class) + 1,110 clean outdoor PlantDoc representations (370 unique images replicated 3x) = **7,710 total records**.

• **Unique Hash Discovery:** The 7,710 records resolve to exactly **6,965 unique image hashes** (6,595 PlantVillage + 370 PlantDoc).

• **Audit of the 5 Duplicate Pairs:** The numerical discrepancy of 5 hashes was traced to upstream Kaggle duplicates with different UUID names. All 5 duplicate pairs share identical class labels, proving zero cross-class conflict:

Duplicate Hash Prefix	Class Category	First Upstream File Name	Second Upstream File Name	Audit Verdict
90220c67...	Tomato Late Blight	348c5e63-9562-4ef8...JPG	d1ea0e62-c0cf-46d2...JPG	Identical label (Zero conflict)
1d202e8c...	Tomato Late Blight	578e93aa-c0f5-46aa...JPG	672e811f-c081-432d...JPG	Identical label (Zero conflict)
7ff8410d...	Tomato Late Blight	7f153a77-33f7-41e1...JPG	b1e102f9-723a-4eb3...JPG	Identical label (Zero conflict)
6e1610d4...	Tomato Late Blight	8f3957eb-30e7-49f8...JPG	81966a01-209a-4c28...JPG	Identical label (Zero conflict)
32e9ffab...	Tomato Healthy	4c0b624b-f288-444a...JPG	a533da56-5fa5-498c...JPG	Identical label (Zero conflict)

Table 5.2: Pure Dual-Domain Training Corpus Distribution

Index	Class Taxonomy Label	Target Crop	Botanical Pathogen / Agent	Unique Hashes	Replicated Total
0	Tomato___Late_blight	Tomato	*Phytophthora infestans* (Oomycete)	1,180	1,368
1	Tomato___Early_blight	Tomato	*Alternaria solani* (Fungus)	1,232	1,518
2	Tomato___healthy	Tomato	Asymptomatic Foliar Tissue	1,166	1,326
3	Potato___Late_blight	Potato	*Phytophthora infestans* (Oomycete)	1,170	1,268
4	Potato___Early_blight	Potato	*Alternaria solani* (Fungus)	1,121	1,280
5	Potato___healthy	Potato	Asymptomatic Foliar Tissue	1,096	950
Total	**Solanaceae Baseline**	**Dual Crop**	**Pure Dual-Domain Corpus**	**6,965**	**7,710**

SECTION 6: COMPREHENSIVE EMPIRICAL EVALUATION MATRICES

Table 6.1: Per-Class Accuracy on 60-Specimen Botanical Benchmark

Foliar Disease Category	Tested	Strict Exact Match	Strict Accuracy (%)	Mean Confidence	Mean Latency (PC)
Potato Early Blight	10	9 / 10	90.0%	99.8%	9.33 ms
Potato Late Blight	10	10 / 10	100.0%	96.1%	8.40 ms
Potato Healthy	10	10 / 10	100.0%	96.2%	6.19 ms
Tomato Early Blight	10	7 / 10	70.0%	85.0%	6.27 ms
Tomato Late Blight	10	7 / 10	70.0%	81.3%	8.04 ms
Tomato Healthy	10	10 / 10	100.0%	100.0%	6.93 ms
Overall / Mean	**60**	**53 / 60**	**88.33% Strict**	**93.1%**	**7.53 ms**

**Note on Pathogen Match: 1 Tomato Late Blight leaf was predicted as Potato Late Blight. In our 6-class model this is marked as an exact classification failure, giving 53/60 (88.33%), but it matches the underlying Phytophthora infestans biological pathogen (54/60, 90.00%).*

Table 6.2: Selected Error Analysis & Real-World Failure Specimens

Specimen Name	Ground Truth	Model Prediction	Confidence	Physical / Optical Root Cause	Engineering Mitigation
healthy_potato_leaves.jpg	Potato Healthy	Potato Late Blight	71.1%	Harsh direct sunlight created sharp drop shadows on lower leaf.	Use 3D optical shroud to block ambient diagonal light.
internet_tea_leaf_ood_01.jpg	Non-Crop (Tea Leaf)	Potato Late Blight	87.0%	Closed-set model forced alien green leaf into 1 of 6 Solanaceae classes.	Future Open-Set Recognition (OSR); entropy check.
V261110-191530...jpg	Tomato Healthy (Night)	Tomato Early Blight	68.4%	Unshielded night torch created specular hotspot with dark rim.	Diffuser disc inside shroud softens illumination.
internet_potato_late_02.jpg	Potato Late Blight	UNCERTAIN	0.0%	Black velvet cloth background triggered peripheral dark void gate.	Safe gating: model correctly avoided guessing.

SECTION 7: PLAIN-LANGUAGE FIRMWARE & SOFTWARE MODULE GUIDE

No hundreds of lines of code. This guide explains how each module functions in plain scientific language:

1. KRISHI_DRISHTI_Controller.ino (Controller Wand Firmware)

Runs on the ESP32 DevKit V1. It initializes the I2C OLED display and DHT11 sensor on GPIO 18. It connects as a Wi-Fi station to the camera's SoftAP. When the user presses SCAN (GPIO 27), it reads temperature and humidity, sends an asynchronous HTTP trigger to the camera, and polls non-blockingly every 250ms for the diagnosis. When the NEXT button (GPIO 26) is pressed, it smoothly rotates between 4 OLED pages.

2. KRISHI_DRISHTI_Camera.ino (Camera Unit & Offline Server)

Runs on the AI-Thinker ESP32-CAM. It configures the OV5640 5MP sensor in UXGA mode (1600×1200) and mounts the 8GB MicroSD in high-speed 1-bit SD_MMC mode. It broadcasts the Wi-Fi SoftAP mesh (192.168.4.1) and hosts an embedded web server in PROGMEM. When `/api/capture` is called, it flashes the torch LED (GPIO 4), captures the frame, writes it to SD storage, and buffers it for inference.

3. MicroclimateTracker.cpp / .h (Agronomic Risk Engine)

An algorithmic library embedded into the controller. It models the Wallin and Hyre potato late blight risk criteria. If Relative Humidity is >= 85% and temperature is between 18°C and 24.5°C, it marks the microclimate as SEVERE RISK. If Relative Humidity is < 85%, it suppresses urgent chemical fungicide recommendations, preventing costly spray misapplications.

4. krishi_ai_daemon.py (Offline Edge-AI Engine)

Runs in Python on the smartphone (or offline browser). It loads the 2.03 MB TFLite model, normalizes captured images, and evaluates the dual-stage gating logic: checking foliar pixel ratios, dark perimeter voids, and Shannon entropy. If the frame is clear and confident, it uploads the diagnosis and targeted organic/chemical advice to `/api/diagnosis/set`.

5. train_model.py (Zero-Leakage Transfer Learning Script)

Implements the two-stage transfer learning training protocol on MobileNetV3-Small. It includes hard cryptographic assertions that verify zero hash overlap with the 60 benchmark leaves and 93 PlantDoc held-out images, ensuring complete scientific integrity.

SECTION 8: GLOSSARY OF CORE SCIENTIFIC & TECHNICAL TERMS

- **Oomycete (Water Mold):** A fungus-like eukaryotic microorganism (kingdom Stramenopila) whose cell walls are composed of cellulose and beta-glucans rather than chitin. Example: Phytophthora infestans (Late Blight).
- **Ascomycete (True Fungus):** A phylum of true fungi characterized by sac-like structures (asci) and chitinous cell walls. Example: Alternaria solani (Early Blight).
- **Plant Disease Triangle:** The fundamental botanical principle stating that infectious disease requires three concurrent factors: a susceptible host, a virulent pathogen, and a favorable microclimate.
- **Edge-AI Computing:** Executing artificial intelligence neural networks locally on edge hardware (microcontrollers, local smartphones) rather than transmitting data to remote cloud servers.
- **MobileNetV3-Small:** An efficient convolutional neural network architecture optimized for edge devices, using depthwise separable convolutions and Squeeze-and-Excitation (SE) attention blocks.
- **Post-Training Quantization (PTQ):** A model compression technique that converts 32-bit floating point weights into 16-bit half-precision floats (Float16), reducing file size by 50% with near-zero accuracy loss.
- **Shannon Information Entropy (H):** A mathematical measure of uncertainty in probability distributions ($H = -\sum p_i \ln p_i$). Used in KRISHI-DRISHTI to detect and reject ambiguous or confused predictions.
- **Out-of-Distribution (OOD):** Images or inputs that do not belong to any of the classes the neural network was trained to recognize (e.g., showing a tea leaf to a tomato/potato classifier).

- **Wi-Fi SoftAP Mesh:** Software-enabled Access Point mode where the ESP32-CAM broadcasts its own local Wi-Fi network (192.168.4.1), allowing the wand and phone to communicate with zero internet.
- **Hyperfocal Sweet Spot:** The fixed optical distance (8–10 cm for the OV5640) where sharpness and depth of field are maximized. Mechanically enforced by the 3D-printed shroud.
- **Wallin & Hyre Epidemiology Models:** Established agricultural forecasting equations that predict Late Blight sporulation based on continuous hours of high relative humidity (> 85%) and temperature (18–24.5°C).